

B Tech 2nd year
MST 1 Computer System Architecture

MM:15

TT:1 hour

Section A is compulsory. Attempt any two from Section B.

Section A

1. ✓ 1. What is Cycle per instruction? How is it calculated?
- ✓ 2. 2. What is the role of the Program counter in the interrupt cycle?
3. 3. Register A holds 8 bit binary value 1011 0001 and Register B holds 01101101 after the execution of following micro operations determine the value of Register A and Register B.
$$A \leftarrow A+B$$
- ✓ 4. 4. What is register transfer language?
- ✓ 5. 5. Write the 16 bit instruction format for I/O reference instructions for mano's computer?

Section B

2. Explain instruction cycle in detail. (5)
3. What is an Addressing mode? Describe various addressing modes in detail. (5)
4. A) Differentiate between structure and function. (2)
- B) How is computer architecture different from computer organization? I/O mechanism an organizational attribute or architectural? (3)

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Department of Computer Science and Engineering

Roll No	MST 1 B Tech 2 nd year	Date: 15.9.25
TT:1 Hr	Computer System Architecture	MM:15
Note: Attempt all questions from Section A and any two from Section B. Write your Roll No. on Question Paper.		
Section A		
Q1	a. What is the role of Program Counter in Interrupt Cycle.	1
	b. What is the difference between direct and indirect address instruction? How many references to memory are required for each type of instruction to bring an operand into a processor register?	2
	c. Define MIPS rate. How MIPS is related with CPI and Throughput.	1
	d. What is Arithmetic shift Left and Arithmetic shift right.	1
Section B		
Q2	Design 4 bit Arithmetic Circuit.	5
Q3	Explain the instruction cycle in detail.	5
Q4	How Computer Architecture is different from Computer Organisation.	5

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Write your Roll No. on Question Paper.

Section A

- Q1 a. What is the Purpose of FGI and FGO flags in Interrupt Cycle.
 b. A digital computer has common bus system for 16 registers of 32 bit each. The bus is constructed with multiplexers.
 i. How many selection inputs are there in each multiplexer?
 ii. How many multiplexers are there in each bus?
 c. Designing of instruction set is an architectural issue or organisational issue. Justify
 d. How many full adders and half adders will be required to add two 32 bit numbers.
 e. What is the Purpose of Program Counter?

5 X

Section B

- Q2) The 8-bit registers AR, BR, CR and DR initially have following values:

A=1110010 B=1111111 C=10111001 D=11101010

Determine the 8-bit values in each register after the execution of following micro operations:

i. $AR \leftarrow AR + BR$ ii. $CR \leftarrow CR \wedge DR, BR \leftarrow BR + 1$ iii. $AR \leftarrow AR - CR$

iv. ashr DR

v. Perform selective clear operation on AR with BR

(5)

- Q3) 4 bit register A contain 7 and 4 bit register B contain -3. Determine the value in resultant registers after AXB (multiply) arithmetic micro- operation with multiplication algorithm. Illustrate all the steps clearly. (5)

- Q4) a) Consider an instruction cycle consisting of fetch, decode, calculate effective address and execute. Explain the purpose of these four phases. (2.5)

- b) Assume that the first instruction is store in location 3F0. Considering PC, IR, AR, DR, AC are Program Counter, Instruction register, Address register, Data Register and Accumulator respectively. Find the values of IR, AR, DR and AC after the fetch cycle. (2.5)

3F0	0FF0	3F0	PC
3F1	8FF!		IR
3F2	1FF2		AR
			DR
FF0	0005		AC
FF!	0007		
FF2			

B. Tech CSE
2nd Year, 3rd Sem (56 and 78 groups)
Subject – Computer System Architecture
MST – 1

Time: 1 Hr.

M. Marks: 15

Note: Section – A is compulsory. Do any two questions from Section – B.

Section – A

- Q1. Describe the significance of Accumulator and Program Counter. (2)
Q2. What do you understand by Memory transfer? (2)
Q3. What is stored program concept? (1)

Section – B

- Q1. Describe the various functions of computer in detail. (5)
Q2. What are micro-operations? Suppose register R2 holds the 8-bit number 11011001. Determine the sequence of binary values in R2 after an arithmetic shift-right micro-operation, followed by two circular shift-right micro-operations and followed by a logical shift-left micro-operation. (5)
Q3. What is instruction cycle? Describe with the help of flowchart. (5)